

GFS SPECIFICATION

**GEOTECHNICAL DESIGN OF SHALLOW FOUNDATIONS
FOR ONSHORE LNG FACILITIES**

GFS IG.34.85.23-Gen.

February 2018

ECCN EAR99

GENERAL FUNCTIONAL STANDARD



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These views are based on the experience acquired during involvement with the design, construction, operation and maintenance of processing units and facilities. Where deemed appropriate GFSs are based on Shell DEPs, international, and industry standards.

The objective is to set the standard for good design and engineering practice to be applied by Shell companies in oil and gas production, oil refining, gas handling, gasification, chemical processing, or any other such facility, and thereby to help achieve maximum technical and economic benefit from standardization.

The information set forth in these publications is provided to Shell companies for their consideration and decision to implement. This is of particular importance where GFSs may not cover every requirement or diversity of condition at each locality.

When Contractors or Manufacturers/Suppliers use GFSs, they shall be solely responsible for such use, including the quality of their work and the attainment of the required designs and engineering standards. In particular, for those requirements not specifically covered, the Principal will typically expect them to follow those design and engineering practices that will achieve at least the same level of integrity as reflected in the GFSs. If in doubt, the Contractor or Manufacturer/Supplier shall, without detracting from its own responsibility, consult the Principal.

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- 1) Operating Units having a Service Agreement with Shell GSI or another Shell Company. The use of GFSs by these Operating Units is subject in all respects to the terms and conditions of the relevant Service Agreement.
- 2) Other parties who are authorised to use GFSs subject to appropriate contractual arrangements (whether as part of a Service Agreement or otherwise).
- 3) Contractors/subcontractors and Manufacturers/Suppliers under a contract with users referred to under 1) or 2) which requires that tenders for projects, materials supplied or - generally - work performed on behalf of the said users comply with the relevant standards.

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All administrative queries should be directed to the Technical Standards Team in Shell GSI.

TABLE OF CONTENTS

1.	INTRODUCTION	4
1.1	SCOPE.....	4
1.2	DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS	4
1.3	DEFINITIONS	4
1.4	CROSS-REFERENCES	6
1.5	SUMMARY OF MAIN CHANGES.....	6
1.6	COMMENTS ON THIS GFS.....	6
1.7	DUAL UNITS.....	6
1.8	NON NORMATIVE TEXT (COMMENTARY).....	6
1.9	UPDATE POLICY	6
2.	GENERAL.....	7
2.1	PURPOSE OF THIS GFS.....	7
2.2	APPLICABLE CODES AND STANDARDS	7
3.	GENERAL REQUIREMENTS	8
3.1	GENERAL.....	8
3.2	SOIL INVESTIGATIONS.....	8
4.	GEOTECHNICAL DESIGN	10
4.1	LOAD AND RESISTANCE FACTOR DESIGN (LRFD).....	10
4.2	GLOBAL FACTOR OF SAFETY (GFoS).....	11
4.3	SETTLEMENT ASSESSMENT.....	13
5.	CONSTRUCTION	14
6.	DOCUMENTATION	15
6.1	GENERAL.....	15
6.2	GEOTECHNICAL DESIGN REPORT.....	15
7.	REFERENCES	16
7.1	SHELL STANDARDS	16
7.2	INDUSTRY STANDARDS	16
7.3	PROJECT STANDARDS	16

1. INTRODUCTION

1.1 SCOPE

This new GFS specifies requirements and gives recommendations for the geotechnical design of shallow foundations.

This GFS is classified as a TIER 2 document. See (1.3) for definitions of TIER 1/TIER 2.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this GFS is confined to Shell companies and, where necessary, to Contractors nominated by them. Any authorized access to GFSs does not for that reason constitute an authorization to any documents, data or information to which the GFSs may refer.

This GFS is intended for use in facilities related to oil and gas production, gas handling oil refining, chemical processing, gasification, distribution and supply/marketing. This GFS may also be applied in other similar facilities.

When GFSs are applied, a Management of Change (MOC) process shall be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this GFS, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and which combination of requirements will be acceptable with regards to the safety, environmental, economic and legal aspects. In all cases the Contractor shall inform the Principal of any deviation from the requirements of this GFS which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned, the objective being to obtain agreement to follow this GFS as closely as possible.

1.3 DEFINITIONS

1.3.1 General definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

TIER 1 packaged equipment solutions are kept evergreen and have a revision cycle.

TIER 2 packaged equipment solutions do not have a revision cycle and updated when needed.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

1.3.2 Specific definitions

Term	Definition
Custodian	Person within the discipline appointed by the GDH to be responsible for the contents, upkeep and approval of a specific packaged equipment solution or other Technical Standard. The Custodian is normally the technical authority on the contents of the document. The Custodian also acts as the focal point for questions relating to the content, meaning and application of the packaged equipment solution.
External Standard	Standard (e.g., ISO, EN, ASME, BS, API) issued under the auspices of a non-Shell standards body.
Management of Change (MoC)	Management of Change (MoC) is the application of management principles to a change situation to prevent unforeseen consequences. The MoC process identifies risks arising from changes and manages these risks before and during execution of the changes.
Project Technical Specification	The specification generated by a project to reflect the specific technical requirements of that project.
Raft Foundation	A shallow foundation system with both its plan dimensions (length and width) approximately 10 times its thickness or greater.
Standard	Document, established by consensus and approved by a recognised body, that provides, for common and repeated use, rules, guidelines or characteristics for activities or their results, aimed at the achievement of the optimum degree of order in a given context.
Strip Foundation	A shallow foundation with length at least 5 times the width and width at least 10 times the thickness.

1.3.3 Abbreviations

Term	Definition
AFC	Approved for Construction
DEP	Design and Engineering Practice
FGL	Finished Ground Level
GDH	Global Discipline Head
GFS	General Functional Standard
GFS	Global Functional Specification
GFoS	Global Factor of Safety
LRFD	Load Resistance Factor Design
O	Operational
OBE	Operating Basis Earthquake

Term	Definition
SSE	Safe Shutdown Earthquake
SSI	Soil Structure Interaction

1.4 CROSS-REFERENCES

Where cross-references to other parts of this GFS are made, the referenced section or clause number is shown in brackets (). Other documents referenced by this GFS are listed in (7).

1.5 SUMMARY OF MAIN CHANGES

This is a new GFS.

1.6 COMMENTS ON THIS GFS

Comments on this GFS may be submitted to the Administrator using the same methods as for submitting comments on DEPs:

Shell DEPs Online (Users with access to Shell DEPs Online)	Enter the Shell DEPs Online system at https://www.shelldeps.com Select a GFS and then go to the details screen for that GFS. Click on the "Give feedback" link, fill in the online form and submit.
DEP Feedback System (Users with access to Shell Wide Web)	Enter comments directly in the DEP Feedback System which is accessible from the Technical Standards Portal http://www.shell.com/standards . Select "Submit DEP Feedback", fill in the online form and submit.
DEP Standard Form (other users)	Use Standard Form PES 00.00.05.82-Gen. to record feedback and email the form to the Administrator at standards@shell.com .

Feedback that has been registered in the Feedback System by using one of the above options will be reviewed by the GFS Custodian for potential improvements to the GFS.

1.7 DUAL UNITS

This GFS contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this GFS indicates text that is non-normative and is provided as explanation or background information only.

Non-normative text is normally indented slightly to the right of the relevant GFS clause.

1.9 UPDATE POLICY

This GFS will be updated when applied in a project in accordance with the framework for TIER 2 Packaged Equipment Solutions.

2. GENERAL

2.1 PURPOSE OF THIS GFS

This GFS provides the requirements for the geotechnical design of foundations including:

- *Pad Foundations (footings);*
- *Strip Foundations;*
- *Raft Foundations.*

2.2 APPLICABLE CODES AND STANDARDS

This GFS is based on American standards and provides minimum requirements for the geotechnical design and construction of shallow foundations.

This document permits the use of both Load Resistance Factor Design (LRFD) and "Global Factor of Safety" (GFoS) philosophies for the determination of geotechnical capacities.

1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.

3. GENERAL REQUIREMENTS

3.1 GENERAL

1. Allowable capacities for both LRFD and GFS methods shall be based on the results of a geotechnical investigation.
 - a. In the absence of project specific documentation settlement limits shall be based on Table 4.3.
2. Foundations shall be designed so that the allowable bearing capacities are not exceeded and differential settlements are minimised.
3. Footing capacity shall be determined considering the highest groundwater table expected over the service life of the structure.
4. Contractor may use LRFD methods for the determination of foundation sizing.
 - a. Contractor shall carry out foundation eccentricity checks using un-factored serviceability load combinations in accordance with Table 14.2 of GFS IG.34.85.02-Gen.
 - b. The eccentricity of the foundation shall be limited to 1/6 and 1/4 respectively when spread footings are founded on soil and rock for serviceability load combinations.
5. Shallow foundations shall be located 300 mm (12 in) below the assessed frost penetration depth.
6. Where required settlements shall be determined using un-factored serviceability load combinations in accordance with Table 14.2 of GFS IG.34.85.02-Gen.
7. Foundation design in arid regions shall consider possibility and impact of expansive clays, reduction in bearing capacity with depth due to soil crust presence, potential collapse settlement and influence of water on ground behaviour.
8. Foundation design in regions of expansive soils shall consider the effect of changes in water effect due to evaporation and infiltration.

3.2 SOIL INVESTIGATIONS

3.2.1 General

1. Soil investigations will be provided by Principal.
2. Unless specified otherwise in the Project Documents:
 - a. Factual data provided shall be "relied upon".
 - b. Interpretative reports shall be for information only.
3. The Contractor shall review the factual reports and verify that the data is sufficient for its design and construction purposes.
 - a. If the Contractor requires additional information on ground conditions, then the Contractor shall carry out an assessment of the scope of the additional work and communicate the same in writing to the Principal at tender stage.
 - b. Contractor shall carry out the additional Site Investigation at its own expense.
4. The final interpretation of the factual data provided to the Contractor and the factual data obtained by the Contractor shall be the Contractor's sole responsibility.

3.2.2 Liquefaction

1. Liquefaction shall be assessed in accordance with GFS IG.34.85.22-Gen.
2. Liquefaction is permitted for systems and structures provided:
 - a. the resulting settlements/lateral displacements are within acceptable limits;
 - b. the supported structures and systems can withstand the imposed deformations and attain the target performance levels;
 - c. the foundation is located below the maximum depth of liquefaction.
3. Soil improvement techniques or piled foundations shall be used to mitigate the effects of liquefaction.

4. GEOTECHNICAL DESIGN**4.1 LOAD AND RESISTANCE FACTOR DESIGN (LRFD)**

1. LRFD strength load factors shall be in accordance with Table 4.1 of GFS IG.34.85.02-Gen.
2. Load factors and combinations for deflection checks and serviceability states shall be in accordance with Table 14.2 of GFS IG.34.85.02-Gen.
3. LRFD strength limit states shall be calculated using average measured values of strength (and stiffness) parameters based on relevant laboratory test data and/or in-situ test data, rather than a lower bound value.
4. LRFD Serviceability limit states and assessments of soil structure interaction (SSI) shall be calculated using both upper and lower bound values of stiffness (and strength) parameters based on relevant laboratory test data and/or in-situ test data.
5. Resistance factors for the geotechnical design of shallow foundations at the strength limit state shall be in accordance with AASHTO as summarised in Table 4.1.

Table 4.1 LRFD Geotechnical Resistance Factors

Basic Load Combination	Resistance Factor		
	Bearing	Sliding ¹	Overall Slope Stability
Normal Operation	AASHTO Table 10.5.5.2.2-1		0.65 AASHTO 11.6.2.3
Earthquake (OBE or DE (2/3xMCE) as per GFS IG.34.85.10)	AASHTO Table 10.5.5.2.2-1		0.65 AASHTO 11.6.2.3
Earthquake (MCE or SSE earthquake as per GFS IG.34.85.10)	1.0, AASHTO10.5.5.3.3		0.9 AASHTO 11.5.8
Blast (Abnormal)	1.0, AASHTO10.5.5.3.3		0.9 AASHTO 11.5.8

Notes:

¹ Passive resistance shall be neglected if the soil providing the passive resistance is poorly placed/compacted or is likely to become soft, loose, disturbed, or where the soil is not compacted against the wall/foundation. Where included in sliding resistance calculations at the strength limit state the resistance factor on the passive resistance shall be taken as 0.5 in accordance with AASHTO Table 10.5.5.2.2-1.

6. The geotechnical design of shallow foundations shall be in accordance with Section 10.6 of AASHTO with the following amendments:
 - a. Modification to 10.6.3.1.3.
 - i. Semi-empirical methods for the design of shallow foundations shall not be used.
 - b. Modification to 10.6.2.6.1.
 - i. The use of presumptive values shall not be used for design.
 7. Shallow foundations located on or near slopes shall be designed in accordance with the following:
 - a. Stability of the slopes as per AASHTO Section 11.6.2.3;
 - b. Influence of slope on foundation design in accordance with AASHTO Section 10.6.3.1.2c;
 - c. Slope stability resistance factors shall be in accordance with Table 4.1.
- 4.2 GLOBAL FACTOR OF SAFETY (GFoS)
1. GFoS load factors shall be in accordance with Table 14.2 of GFS IG.34.85.02-Gen.
 2. Load factors and combinations for deflection checks and serviceability states shall be in accordance with Table 14.2 of GFS IG.34.85.02-Gen.
 3. Characteristic geotechnical parameters shall be selected for the determination of capacities under the GFoS approach.
 - a. Definition of characteristic value may utilise definition as per EN1997-1.
 4. The selection of characteristic values for geotechnical parameters shall be based results and derived values from laboratory and field tests.
 5. The selection of characteristic values for geotechnical parameters shall take account of the following:
 - a. geological and other background information, such as data from previous projects;
 - b. the variability of the measured property values and other relevant information, e.g., from existing knowledge;
 - c. the extent of the field and laboratory investigation;
 - d. the type and number of samples;
 - e. the extent of the zone of ground governing the behaviour of the geotechnical structure at the limit state being considered.
 6. The most unfavourable combination of lower and upper values of independent parameters shall be used.
 7. The geotechnical resistance of shallow foundations shall be in accordance with section 10.6 of AASHTO with equation 1.3.2.1-1 replaced by the following:

$$\sum \eta_i \gamma_i Q_i \leq \frac{R_n}{FoS} \quad \text{Eqn. 1}$$

Where, Factor of Safety (FoS) is defined in Table 4.2. η_i , λ_i , Q_i and R_n are as per AASHTO.

Table 4.2 Global Factors of Safety (FoS) for Shallow Foundations

Loading Condition	Loading Phase				
	Normal Operation	Hydrotest	Earthquake		Abnormal
			Operational (OBE or DE (2/3xMCE) as per GFS IG.34.85.10)	Life Safety (MCE or SSE earthquake as per GFS IG.34.85.10)	
Bearing	3.0	2.4	2.25	1.5	1.5
Sliding	1.5	-	1.5	1.2	1.2
Slope Stability	1.5	1.5	1.5	1.15	1.15
Buoyancy	1.2	-	-	-	-
Notes: 1. Passive resistance shall be neglected if the soil providing the passive resistance is poorly placed/compacted or is likely to become soft, loose, disturbed, or where the soil is not compacted against the wall/foundation. Where included in sliding resistance calculations at the strength limit state the resistance factor on the passive resistance shall be taken as 0.5 in accordance with AASHTO Table 10.5.5.2.2-1.					

8. The geotechnical design of shallow foundations shall be in accordance with Section 10.6 of AASHTO with the following amendments:
 - a. Modification to 10.6.3.1.3.
 - i. Semi-empirical methods for the design of shallow foundations shall not be used.
 - b. Modification to 10.6.2.6.1.
 - i. The use of presumptive values shall not be used for design.
9. Shallow foundations located on or near slopes shall be designed in accordance with the following
 - a. Stability of the slopes as per AASHTO Section 11.6.2.3;
 - b. Influence of slope on foundation design in accordance with AASHTO Section 10.6.3.1.2c;
 - c. Slope stability resistance factors shall be in accordance with Table 4.1.

4.3 SETTLEMENT ASSESSMENT

1. Settlement limits for total and differential settlements shall be defined for the foundation by the Contractor responsible for the design of the superstructure.
2. Settlement calculations shall be in accordance with Section 10.6.2 of AASHTO with the following amendments:
 - a. Modification to 10.6.2.4.3 of AASHTO.
 - i. Immediate or elastic settlement of footings founded on cohesive soils can be estimated using Eq.10.6.2.4.2-1 of AASHTO with the appropriate value of the soil modulus.
3. For adjacent foundations where the pressure bulbs due to applied foundation loads overlap, the influence of adjacent footings on settlements shall be considered for foundation design.
4. Unless specified in the geotechnical report settlement limits shall be as per taken as Table 4.3.

Table 4.3 Settlement Limits

Foundation Type	Settlement	
	Maximum Total	Maximum Differential
Foundation in unit areas, utility areas and foundations for Plant buildings.	25mm (1 in)	20mm (0.8 in)
Non Plant Buildings	40mm (1.6 in)	20mm (0.8 in)
Raft Foundations	40mm (1.6 in)	20mm (0.8 in)

5. CONSTRUCTION

1. The following construction aspects shall be detailed in the “Geotechnical Design Report” and annotated on the Approved for Construction (AFC) Foundation Drawings:
 - a. the bearing stratum considered in design for supporting the foundation, the minimum embedment of the foundation in the bearing stratum and expected elevation of the contact surface;
 - b. in-situ tests required to confirm design assumptions and acceptance criteria;
 - c. in cases where the bearing layer may be susceptible to rapid softening, the maximum time between excavation and concreting;
 - d. where excavations deeper than 1.2 m (4 ft) are required to reach the foundation level, definition of safe slope geometry (or any bracing/temporary structure) as demonstrated by specific calculations;
 - e. groundwater control methods and schemes which do not adversely affect the characteristics of the bearing stratum (for example by causing heave or settlement) or the performance of adjacent foundations;
 - f. backfilling of excavated pits shall be in accordance with GFS IG.34.85.22-Gen.

6. DOCUMENTATION**6.1 GENERAL**

1. Documentation shall be submitted for review or acceptance by Principal in conformance to Table 6.1.

Table 6.1 Deliverable Requirements

Deliverable Type	Deliverable Title	Principal Action	Comments
Documents to be Submitted with Technical Bid			
Note	Requirements for additional soil investigation	ACCEPT	As per (3.2)
Design	Design Basis	ACCEPT	Include: Project-specific variation from this GFS Method of analysis and software for analysing
Design, Method Statement, ITPs and HSE/Quality Reports			
Design	Geotechnical design report	ACCEPT	As per (6.2)
Design	Construction drawings	REVIEW	Include: Site-specific construction process affecting foundation design

6.2 GEOTECHNICAL DESIGN REPORT

1. Contractor shall include the following details in the Geotechnical Design Report:
 - a. site condition and design soil parameters for foundation design;
 - b. settlement limits (short term and long-term combined) considered for the analysis including analyses of loading over the construction period including any hydro-testing;
 - c. Ultimate Bearing Capacities for different footing sizes at varying depths;
 - d. Design Bearing Capacities (LRFD or GFS) to meet allowable capacity and settlement limits for different footing sizes and founding depths;
 - e. Minimum embedment below FGL (Finished Ground Level);
 - f. safe slope configuration for excavations;
 - g. site-specific construction process affecting foundation design;
 - h. detailed calculations and assumptions for arriving at the design bearing capacities.

7. REFERENCES

In this GFS, reference is made to the following publications:

- NOTES:
1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.
 2. The referenced external standards are available to Shell staff on the SWW (Shell Wide Web) at <http://swww.shell.com/standards/>.

7.1 SHELL STANDARDS**SHELL STANDARDS**

Procedure for packaged equipment solutions publications	PES 00.00.00.31-Gen.
Packaged equipment solutions feedback form	PES 00.00.05.82-Gen.
Loading for onshore LNG facilities	GFS IG.34.85.02-Gen.
Seismic design for onshore LNG facilities	GFS IG.34.85.10-Gen.
Loading Specification for Onshore LNG Facilities	GFS IG.34.85.23-Gen.
Site Preparation and Earthworks for Onshore LNG Facilities	GFS IG.34.85.22-Gen.
Seismic Design for Onshore LNG facilities	GFS IG.34.85.24-Gen.

7.2 INDUSTRY STANDARDS**AMERICAN STANDARDS**

AASHTO LRFD Bridge Design Specifications (7 th Edition)	AASHTO
Eurocode 7: Geotechnical design - Part 1: General rules	EN 1997-1

7.3 PROJECT STANDARDS

1. The project and Engineering Contractor shall apply the standards as been specified above and no project specific technical standards and/or specifications shall be made on top of these.

It is recognized that the Engineering Contractor has its own procedures and standards to manage a project, from that perspective it is allowed to develop a limited number of project specific standards, specifications, procedures that are required to manage and administer the project execution.